

Saverio Mattia Merenda

MASTER'S STUDENT

Reggio Emilia - Parma, Italy

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Education

University of Parma

M.S. IN COMPUTER SCIENCE, EXPECTED GRADUATION YEAR: JULY 2026

Parma, Italy

From Sep. 2024

- Topics: **Software Verification, Machine Learning & AI.**
- Best Marks (so far): **Compilers, Quantum Computing, Big Data & Data Mining, AI.**

University of Parma

B.S. IN COMPUTER SCIENCE, GRADE: 108/110

Parma, Italy

Sep. 2021 - Jul. 2024

- Thesis Title: *Construction of complete Control-Flow Graphs for bytecode EVM.*
- Best Marks: **Calculus, Programming, Algorithms & DS, Cloud Administration, Database, AI, HPC.**
- Got different ERGO scholarships given to promising Unipr students.

Experience

University of Parma

RESEARCHER AND SOFTWARE DEVELOPER

Parma, Italy

From Jan. 2025

- **Development of xEVMLiSA:** Currently working on the early stages of xEVMLiSA, a project aimed at expanding EVMLiSA to analyze smart contracts across multiple blockchains (cross-chain).

University of Parma

TEACHING ASSISTANT AND TUTOR

Parma, Italy

Sep. 2024 - Dec. 2025

- Teaching Assistant and Tutor for the "Fondamenti di Programmazione A + B (15 CFU)" bachelor course in the Computer Science degree.

University of Parma - Internship

RESEARCHER AND SOFTWARE DEVELOPER

Parma, Italy

Sep. 2023 - May 2025

- Contributed to the **development of EVMLiSA**, a static analyzer for Ethereum smart contracts built on the LiSA (Library for Static Analysis) framework, using abstract interpretation to construct sound Control-Flow Graphs (CFGs).
- **Designed and implemented** multiple security checkers to detect common vulnerabilities such as reentrancy, randomness dependency, enhancing the analyzer's precision and robustness.
- Applied **advanced design patterns** and **software architecture principles** to improve the scalability and maintainability of the tool.
- The project led to two peer-reviewed publications: the workshop paper *FTfJP@ISSTA/ECOOP (2024)* and the journal article *Blockchain: Research and Applications (2025)*.

HeliopsyLab

SOFTWARE DEVELOPER

Reggio Emilia, Italy

Jan. 2021 - Sep. 2021

- Developed and implemented cutting-edge management software for hospital facilities in the Lazio and Lombardia regions, streamlining operations and reducing administrative inefficiencies by 15%.
- Advanced new features within the Picasso and Mirth integration software, enhancing its capabilities and functionality.

Publications

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|------|---|----------------|
| 2025 | Blockchain: Research and Applications , V. Arceri, <u>S. M. Merenda</u> , L. Negrini, L. Olivieri, E. Zaffanella:
"EVMLiSA: Sound Static Control-Flow Graph Construction for EVM Bytecode". (doi: 10.1016/j.bcra.2025.100384) | Journal paper |
| 2024 | FTfJP@ISSTA/ECOOP 2024 , V. Arceri, <u>S. M. Merenda</u> , G. Dolcetti, L. Negrini, L. Olivieri, E. Zaffanella:
"Towards a Sound Construction of EVM Bytecode Control-flow Graphs". (doi: 10.1145/3678721.3686227) | Workshop paper |

Talks

INTERNATIONAL CONFERANCES AND WORKSHOP

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|------|---|-----------------|
| 2024 | Towards a Sound Construction of EVM Bytecode Control-flow Graphs , 26th International Workshop on Formal Techniques for Java-like Programs, FTfJP 2024. [slides] | Vienna, Austria |
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Research Projects

Research Participant, "LLMs Meet Static Analysis: improving quality and reliability of AI-generated code"

Parma, Italy

ISCRA PROJECT (CLASS C), CINECA

2024

The goal of the project is to conduct an extensive quality and safety evaluation of the code generated with some of the most popular and open-source LLMs employing static analyzers, that can detect vulnerabilities and run-time errors statically, without executing the code. Once this information is available, it will be included in the code-generation task, to guide the LLM itself to produce a more precise and safe output, in which static analysis is somehow introduced in the pipeline of the code-generation task.

Tools and Software

EVMLiSA: an abstract interpretation-based static analyzer for EVM bytecode

JAVA, GRADLE

From Sep. 2023

EVMLiSA is a specific implementation of a static analyzer using the LiSA (Library for Static Analysis) library to conduct static analysis of Ethereum Virtual Machine (EVM) bytecode. In particular, it is dedicated to generate a sound Control-Flow Graph (CFG) of smart contracts deployed on the Ethereum blockchain. EVMLiSA's primary objective is to provide semantic information and valuable warnings for developers and security auditors. EVMLiSA is distributed under the MIT license, and it is available on GitHub: github.com/lisa-analyzer/evm-lisa.

Events

2025 **Conference participant**, CSV 2025, 4rd Challenges of Software Verification Symposium 2025. [\[link\]](#)

Venice, Italy

2024 **Student**, Lipari Summer School on Abstract Interpretation. [\[link\]](#)

Lipari, Italy

2024 **Conference participant**, CSV 2024, 3rd Challenges of Software Verification Symposium 2024. [\[link\]](#)

Venice, Italy

Academic Activity

Personal Financial AI Agent

PYTHON, OLLAMA, OPENAI, GOOGLE GEMINI

2025

- Developed a system of interacting **AI agents** acting as a **personal financial advisor**, capable of providing multilingual financial guidance and portfolio analysis.
- Integrated multiple **LLM providers** (Ollama, Google Gemini, OpenAI) for flexible online or fully offline operation.
- Implemented **interactive conversations**, **financial profile extraction**, and **portfolio generation** through a web interface.
- Gained experience in **LLM orchestration**, **Python backend design**, and the integration of conversational AI with financial data analysis.
- Source Code: github.com/merendamattia/personal-financial-ai-agent

Isolette: Interactive Scenario Visualization and UI Simulation

C, VUE.JS, TYPESCRIPT

2025

- Contributed to the **Isolette** project, a safety-critical system simulating a neonatal incubator that monitors and controls temperature.
- Designed and implemented the **User Interface Hardware** (UIH) and its simulator (UIHS) to ensure behavioral consistency during testing.
- Developed an **interactive web interface** using Vue and TypeScript with real-time temperature charts, timeline navigation, and JSON-based validation.
- Built a **Scenario Visualization Framework** replacing gnuplot, using Vue and Chart.js for dynamic, browser-based data visualization.
- The result is a **scalable, browser-based visualization frontend** that improves usability, speeds up development, and removes external dependencies.

Impact of Data Preprocessing on Neural Network Performance

PYTHON

2025

- Investigated how **data preprocessing techniques**—including NaN handling, outlier removal, normalization, and feature transformation—affect the accuracy and stability of **feed-forward neural networks**.
- Implemented multiple preprocessing pipelines and benchmarked them on diverse **classification and regression** datasets.
- Gained strong experience in **data quality analysis**, **model evaluation**, and **experimental methodology**.
- Source Code: github.com/merendamattia/neural-network-performance-by-data-quality

GraphQL API Design and Comparative Analysis

GRAPHQL, REST

2025

- Designed and implemented a **GraphQL API**, defining schema, queries, and mutations for flexible and efficient data retrieval.
- Compared **GraphQL** and **REST** architectures, analyzing efficiency in terms of data transfer, query complexity, and latency.
- Highlighted GraphQL's benefits in reducing over-fetching and under-fetching, improving client-side data control.
- Strengthened skills in **API design**, **backend integration**, and **data modeling**.

Quantum Portfolio Optimization

PYTHON

2025

- Explored **quantum algorithms** for portfolio optimization using **QAOA** and **VQE**.
- Formulated the optimization problem as a **Quadratic Unconstrained Binary Optimization (QUBO)** model and implemented simulations using **Qiskit**.
- Compared performance under ideal and noisy conditions, discussing the potential of **quantum computing in finance** and its current hardware limitations.
- Source Code: github.com/merendamattia/quantum-portfolio-optimization

Optimization of Academic Guarantors: a Declarative Approach

ASP, PYTHON

2024

- Developed an **automated system** for assigning academic guarantors to university courses, ensuring compliance with ministerial regulations.
- Designed a **data preprocessing pipeline** and encoded the optimization logic in **Answer Set Programming (ASP)** to balance resource allocation.
- Demonstrated the effectiveness of **declarative programming** in solving complex institutional optimization problems.
- Source Code: github.com/merendamattia/ottimizzazione-garanti-accademici

Deep Neural Network

C++

2024

- Developed a **neural network framework** supporting customizable architectures, activation functions, and training via **backpropagation** and **gradient descent**.
- Implemented **model serialization** and loading to facilitate retraining and experiment reproducibility.
- Gained solid experience with **machine learning fundamentals**, **optimization algorithms**, and **C++ object-oriented design**.
- Source Code: github.com/merendamattia/deep-neural-network

Extracurricular Activity

My-gpt4

PYTHON

2023

- Crafted a cutting-edge Python integration leveraging multiple reverse-engineered language-model APIs, contributing to the decentralization of the AI industry.
- Source Code: github.com/merendamattia/my-gpt4

Tracking Messages on Bitcoin Blockchain using OP-RETURN field

JS, HTML

2022

- Devised an innovative algorithm to securely trace immutably written messages within the Bitcoin blockchain, utilizing transaction hashes for enhanced transparency.
- Demo: merendamattia.com/dev/btc/tracking/
- Source Code: github.com/merendamattia/op-return-tracking-message-bitcoin

Development of Various Bots

PYTHON, JS

From 2021

- Developed a sophisticated bot designed to navigate financial markets, employing strategic approaches to optimize profits while minimizing potential losses.
- Customized Telegram bots designed for personal optimization, streamlining and enhancing my daily routine.